

Airbnb Dynamic Pricing Recommendation Engine

Data Analytics & Machine Learning Internship Project

1. Abstract

This project develops a data-driven Airbnb pricing recommendation engine using New York City Airbnb Open Data (2019). The workflow combines data cleaning, exploratory analysis, machine learning, and an interactive Streamlit dashboard. A Random Forest Regression model uses listing characteristics to recommend a nightly price. The dashboard allows users to experiment with listing attributes and view market-level pricing patterns.

2. Introduction

Airbnb prices vary substantially across locations, room types, and listing characteristics. The objective of this project is to understand these pricing patterns and build a practical pricing recommendation tool. The project follows an end-to-end analytics workflow: dataset preparation, exploratory data analysis, feature preparation, model training and evaluation, and interactive visualization.

3. Dataset & Data Preparation

The project uses the New York City Airbnb Open Data (2019) dataset. The original dataset contains 48,895 listings and 16 attributes. For pricing analysis, listings with a price of \$0 and listings above \$1,000 were removed, leaving 48,645 listings. Missing values in **reviews_per_month** were treated as zero for the model because they represent listings without recorded monthly reviews in this dataset.

4. Exploratory Data Analysis

- Average price was compared across neighbourhood groups and room types.
- Minimum-night patterns and neighbourhood-level price variation were examined.
- Listing counts were analyzed to identify neighbourhoods with high listing volume.
- Manhattan had the highest average price among the five neighbourhood groups.
- Entire home/apartment listings had a higher average price than private and shared rooms.

5. Machine Learning Approach

A **Random Forest Regression** model was trained to recommend listing prices. The input features were neighbourhood group, neighbourhood, room type, minimum nights, number of reviews, reviews per month, calculated host listings count, and availability. Categorical variables were converted using one-hot encoding. The data was divided into training and testing sets using an 80/20 split.

Evaluation Metric	Result
Mean Absolute Error (MAE)	\$50.10
Root Mean Squared Error (RMSE)	\$89.37
R ² Score	0.43

The model explains approximately 43% of the variation in listing prices. It is therefore presented as a pricing recommendation tool rather than an exact price predictor.

6. Interactive Dashboard

A Streamlit dashboard was developed as the user-facing interface. Users can select neighbourhood group, neighbourhood, room type, minimum nights, number of reviews, reviews per month, host listing count, and availability. The dashboard then generates a machine-learning-based recommended nightly price and displays market overview metrics and price comparisons.

7. Tools & Technologies

Python • Pandas • NumPy • Scikit-learn • Random Forest Regression • Joblib • Streamlit

8. Conclusion

The project demonstrates how data analytics and machine learning can be combined to support Airbnb pricing decisions. Exploratory analysis reveals meaningful differences in prices across locations and room types, while the Random Forest model provides a data-driven recommendation mechanism. The Streamlit dashboard makes the analysis interactive and easier to use, resulting in an end-to-end pricing analytics solution.

Project deliverables: Python analysis and pricing engine • trained model files • interactive dashboard • project documentation